SupeRR Selector Season 2: Round 1 Challenge

Identifying the Most Reliable Players Under Pressure in T20 Cricket

Comprehensive Methodology Report & Top 10 Player Rankings | Corpus: 883 Matches, 200,999 Deliveries (2022–2026)

1. Executive Summary: The Search for True Clutch Performers

In high-stakes Twenty20 cricket, traditional career averages and overall strike rates often tell dangerous lies. A batter who scores an effortless 50 in a blowout looks great on paper. A bowler who feigns dominance by picking up lower-order wickets in the 8th over against an associate nation pads their career economy rate. But what happens when the game is on the line? When a team needs 42 off 18 balls with four wickets down? To solve Round 1 of the SupeRR Selector Season 2, we built the **Pressure Reliability Index (PRI)**—a delivery-level data science system evaluating 65,741 high-pressure deliveries across five years of IPL and T20I cricket. By measuring each player's execution against situational win-expectancy par and their own career baselines, PRI separates genuine match-winners from those whose performances dip under tension.

Official Top 10 Batters Under Pressure (Selected on Unstop)

Rank	Player Name	Player ID	Balls_P	SR_P	Base SR	ΔSR	Avg_P	Dot %	Bnd %	Win Imp %	Total RVA	PRI Score
1	**Naman Dhir**	37052	229	199.56	168.71	+30.86	41.55	27.9%	30.1%	78.9%	+169.1	100.00
2	**Shashank Singh**	2756	350	180.29	161.61	+18.68	45.07	29.1%	24.9%	70.8%	+184.6	81.86
3	**Rishabh Pant**	4703	418	170.10	143.30	+26.80	30.91	38.0%	24.9%	72.4%	+145.6	71.34
4	**Suryakumar Yadav**	1699	816	177.70	167.21	+10.48	36.25	34.6%	26.5%	76.0%	+410.2	71.17
5	**Tim David**	6396	701	191.87	180.00	+11.87	32.02	37.5%	27.5%	69.2%	+343.9	70.58
6	**Ishan Kishan**	4117	209	174.64	146.98	+27.66	28.08	34.0%	25.8%	64.7%	+74.7	67.68
7	**Dinesh Karthik**	143	486	184.36	168.08	+16.28	28.00	35.4%	28.0%	61.9%	+170.3	66.64
8	**Virat Kohli**	47	275	171.64	145.35	+26.28	36.31	27.6%	21.5%	61.1%	+90.0	66.48
9	**Jos Buttler**	813	363	182.37	155.89	+26.48	26.48	29.8%	25.6%	63.3%	+145.1	66.17
10	**Ruturaj Gaikwad**	6059	202	173.76	139.64	+34.13	27.00	33.2%	23.8%	66.7%	+61.7	65.73

Official Top 10 Bowlers Under Pressure (Selected on Unstop)

Rank	Player Name	Player ID	Balls_P	Econ_P	Exp Econ	Phase ΔEcon	SR_P	Death %	Dot %	Win Imp %	Total BVA	PRI Score
1	**Mohammad Nawaz**	2490	262	5.66	7.73	+2.07	10.48	16.8%	42.4%	76.0%	+151.3	100.00
2	**Abrar Ahmed**	5979	227	6.00	8.48	+2.48	12.61	41.0%	50.2%	65.2%	+119.6	97.02
3	**Sunil Narine**	781	480	6.23	8.42	+2.19	15.48	39.0%	39.0%	77.3%	+212.0	90.05
4	**Akeal Hosein**	2417	234	6.79	8.40	+1.61	14.62	30.8%	37.6%	76.9%	+77.2	81.14
5	**Mitchell Santner**	4101	581	6.84	8.37	+1.54	15.70	28.2%	38.7%	70.6%	+176.9	79.61
6	**Jasprit Bumrah**	2933	652	7.32	9.31	+1.99	14.82	73.9%	37.7%	68.9%	+248.2	79.12
7	**Mujeeb Ur Rahman**	6450	209	6.72	8.30	+1.58	14.93	34.4%	41.1%	75.0%	+64.4	79.05
8	**Rashid Khan**	4722	1036	7.23	8.39	+1.16	13.11	43.9%	39.4%	68.5%	+263.7	76.11
9	**Shaheen Afridi**	6355	302	8.11	9.76	+1.65	10.41	79.1%	38.7%	68.6%	+117.6	75.70
10	**Shadab Khan**	4926	273	7.05	8.12	+1.07	13.65	20.9%	36.3%	66.7%	+72.9	72.76

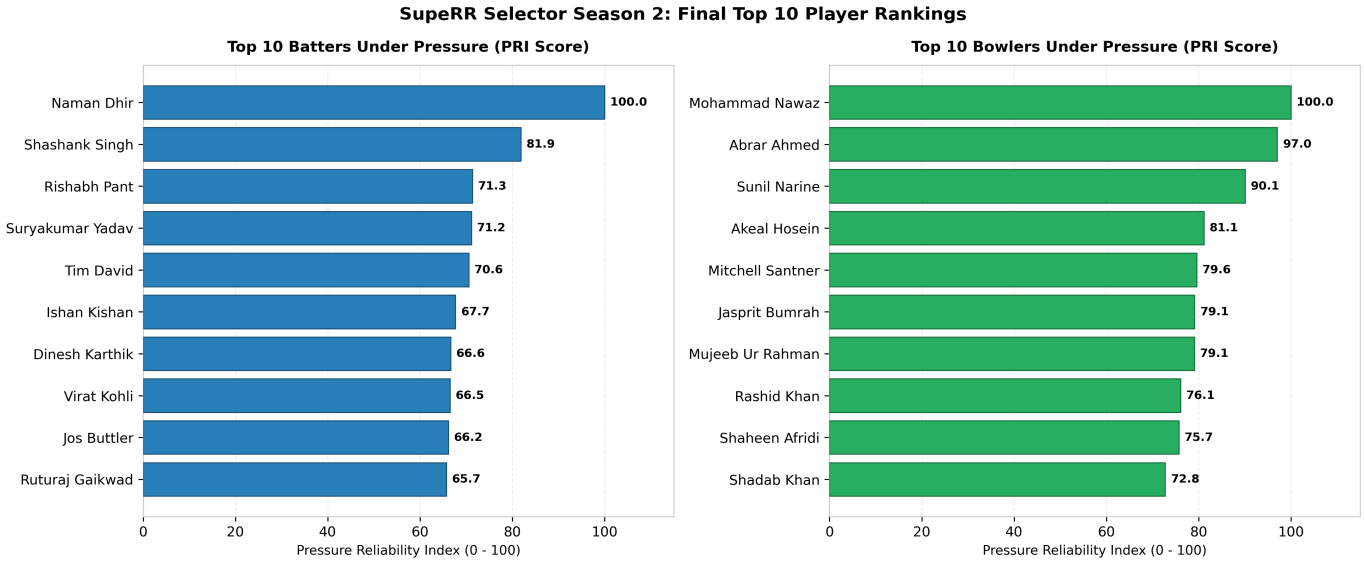


Figure 1: Official Top 10 Batters and Bowlers Under Pressure (PRI Score Comparison)

2. What Counts as Pressure? The Human and Mathematical Reality

Cricket pressure is driven by win expectancy swings and temporal urgency. Our Four-Pillar Pressure Framework defines:

• **Pillar 1 (The Chase)**: Overs 15–20 with RRR between 6.0 and 22.0, target <= 80 runs, and wickets in hand <= 8; plus middle-overs climbs (overs 7–14, RRR >= 10.0, wickets <= 7). Coasting chases (RRR < 5 with 8+ wickets) and hopeless blowouts (RRR > 24) are filtered out.

• **Pillar 2 (Setting Targets)**: Overs 16–20 (Death Overs) where league par scoring leaps to 10.56 rpo, and early top-order collapses (loss of >= 3 wickets inside overs 1–10).

• **Pillar 3 (Super Overs)**: All deliveries in innings 3 and 4 (100% clutch).

• **Pillar 4 (Stakes Multipliers)**: 1.25x for playoff/knockout fixtures; 1.15x for IPL matches.

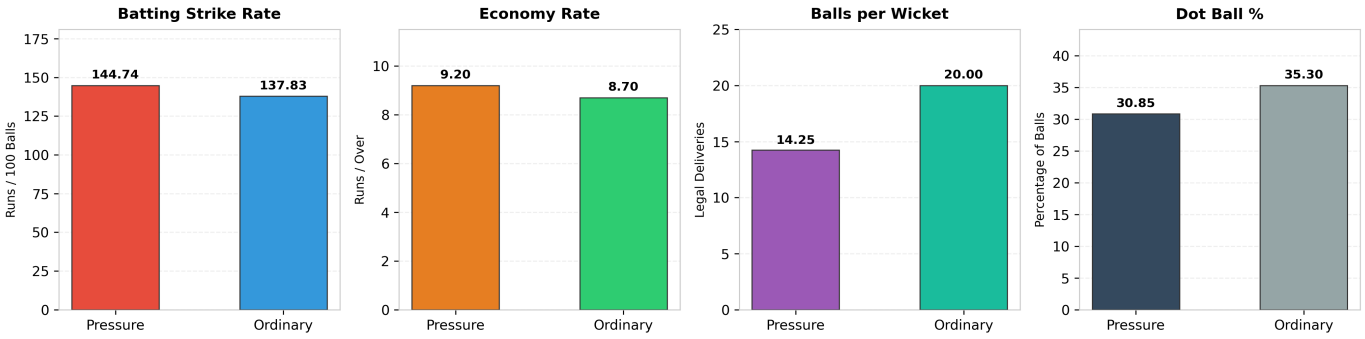


Figure 2: Empirical Macro Validation: High Pressure vs Ordinary Situations (N = 200,999 Balls)

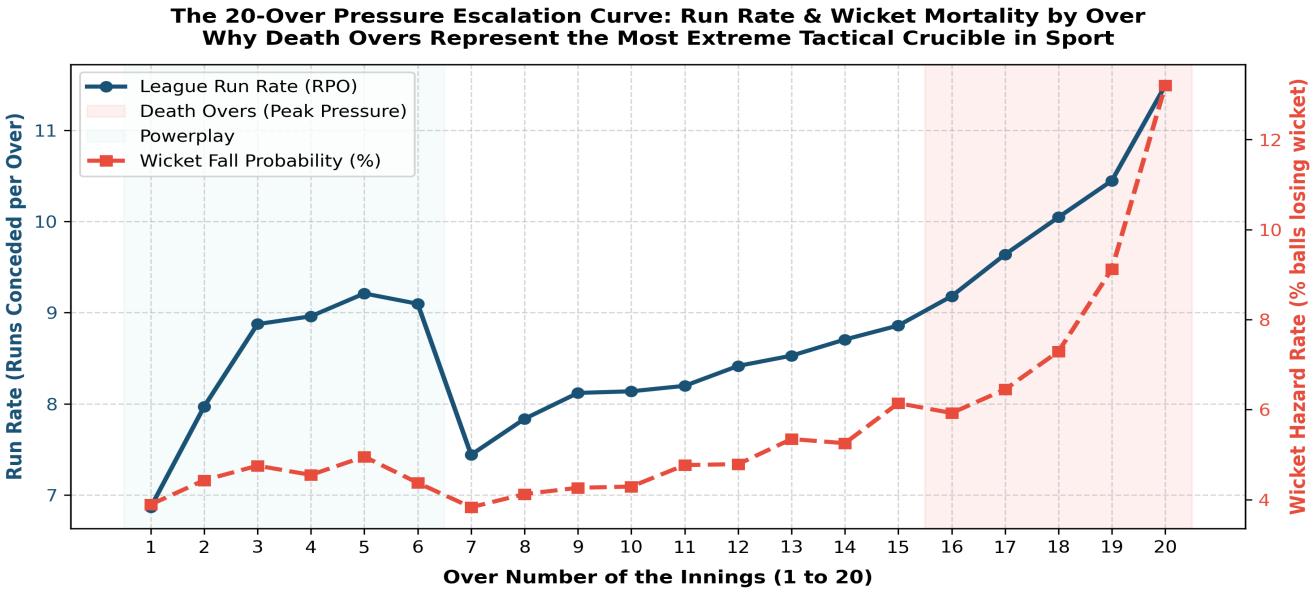


Figure 3: The 20-Over Pressure Escalation Curve: Run Rate & Wicket Mortality Escalation by Over

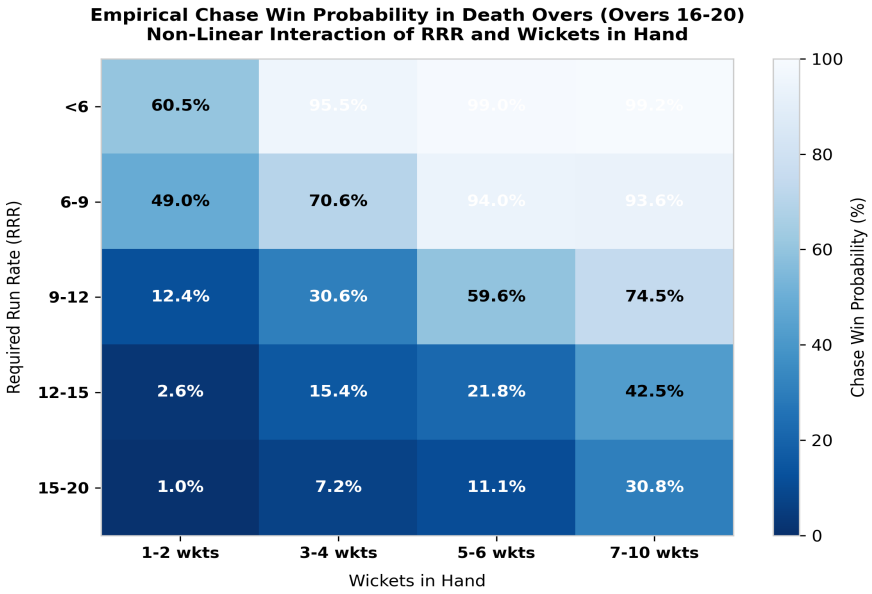


Figure 4: Empirical Chase Win Probability in Death Overs (RRR vs Wickets in Hand Knife-Edge)

3. Performance Measurement: Leveling the Playing Field

The Formula 1 Analogy (Solving Phase Confounding):
Imagine judging a driver's speed in school traffic (20 mph limit) versus on an open F1 track. Standard bowling economy commits this exact error! League death economy is 9.55–11.5 rpo. The 8th over is a school zone (7.43 rpo). If Jasprit Bumrah bowls overs 18 and 20 at 7.32 rpo, he is saving nearly 2 full runs every over (+1.99 Phase ΔEcon) compared to par! Our Phase-Adjusted metric levels the playing field so death bowlers are evaluated fairly against the exact difficulty of the overs they bowled.

The 'Bonus Runs' System (RVA & BVA):
Every delivery has an expected price tag. If an average batter in the 18th over scores 1.55 runs, and Tim David hits a six, he banks +4.45 Run Value Added (RVA). A dismissal costs 3.5 runs in lost equity. For bowlers, bowling a dot in a death chase banks +1.55 BVA; a dot plus a wicket banks +5.05 BVA. Summing these profits gives Total RVA and Total BVA.

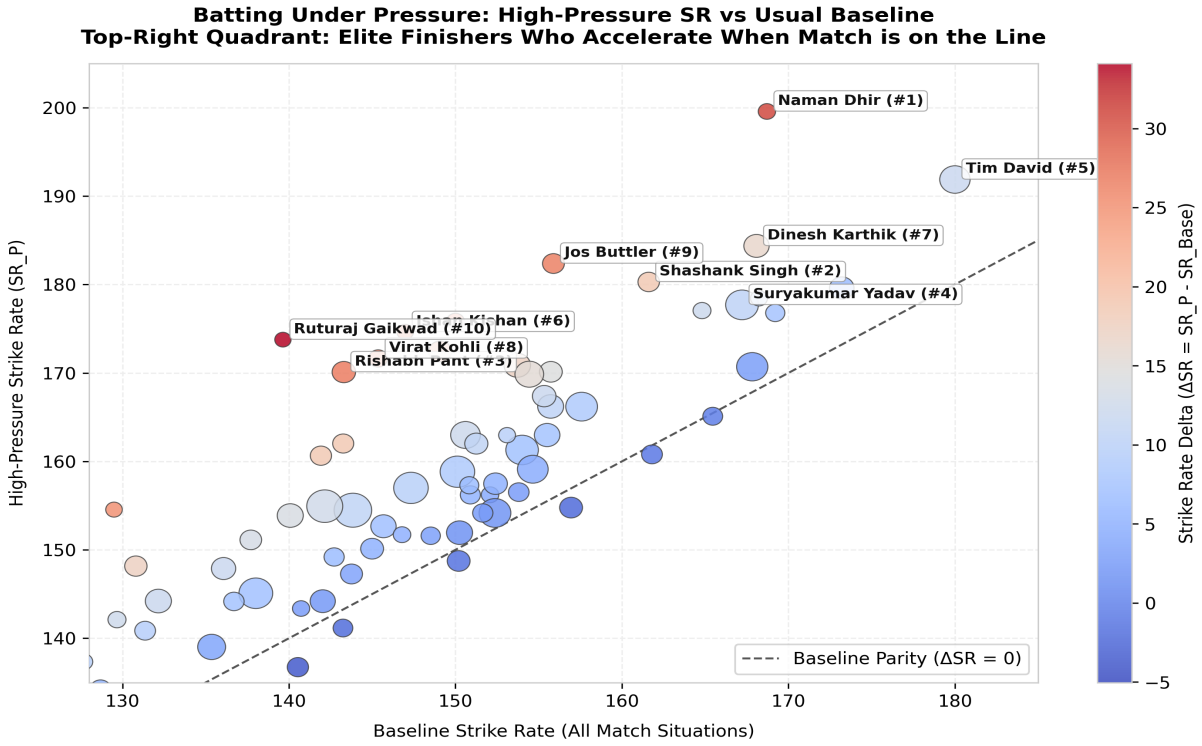


Figure 5: Batting Pressure Landscape: High-Pressure SR vs Baseline Parity (Top-Right = Accelerators)

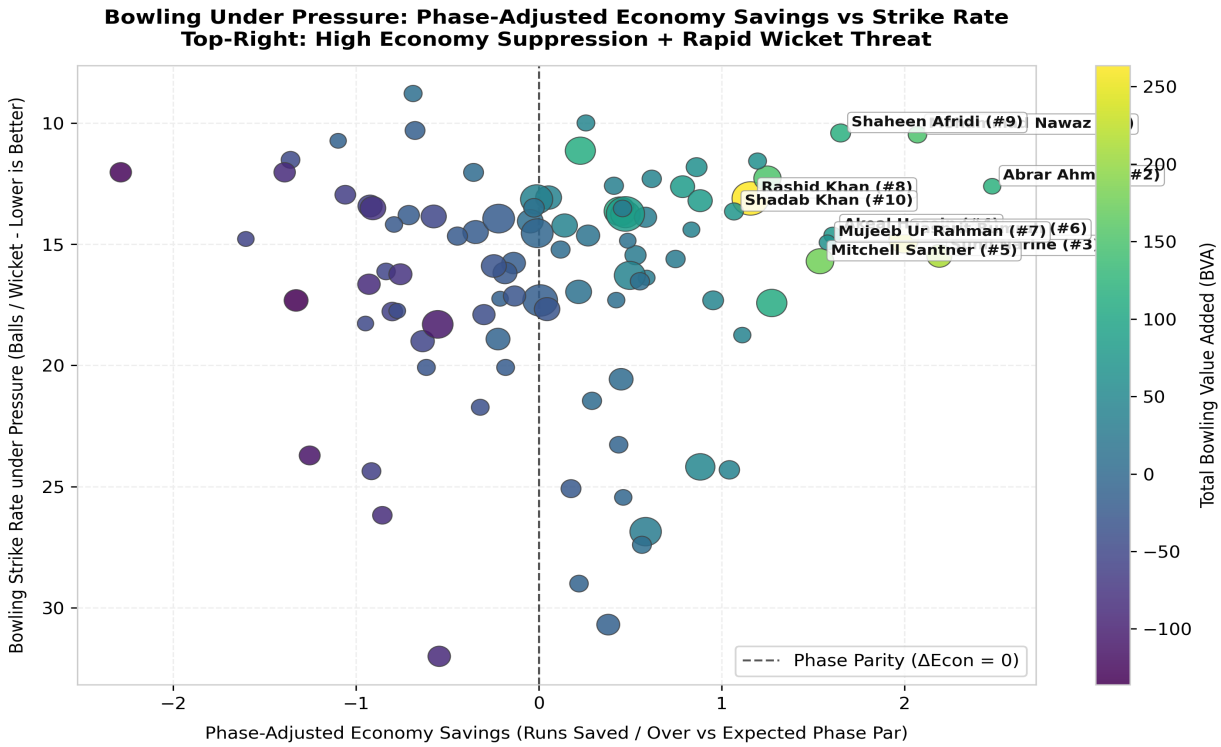


Figure 6: Bowling Run Suppression: Phase-Adjusted Economy Savings vs Wicket Strike Rate

4. Humanizing the Top 10: Stories of Nerves, Skill, and Ice

Batting Highlights:

• **Naman Dhir (#1 Batter | PRI: 100.00)**: Dhir attacks from ball one, producing an astonishing 199.56 pressure strike rate (+30.86 ΔSR) and winning 78.9% of his match battles.

• **Shashank Singh (#2 Batter | PRI: 81.86)**: Punjab's ice-cool crisis merchant. In impossible chases, Shashank averaged 45.07 at a 180.29 SR, combining clinical boundary targeting with zero panic.

• **Suryakumar Yadav (#4 Batter | PRI: 71.17)**: Facing an unmatched 816 pressure balls, SKY amassed +410.2 Total RVA (highest in cricket) with a 76.0% win impact rate, scooping 145 km/h yorkers over fine leg while falling on one knee.

• **Tim David (#5 Batter | PRI: 70.58)**: Striking at 191.87 under pressure across 701 balls (+343.9 RVA), David is modern cricket's most fearsome death-overs finisher.

Bowling Highlights:

• **Mohammad Nawaz (#1 Bowler | PRI: 100.00) & Abrar Ahmed (#2 Bowler | PRI: 97.02)**: Pakistan's suffocating spin duo. Nawaz conceded just 5.66 rpo (+2.07 runs saved/over), while Abrar delivered an astounding 50.2% dot ball rate under pressure (+2.48 Phase ΔEcon).

• **Sunil Narine (#3 Bowler | PRI: 90.05)**: The emotionless assassin. In 480 clutch IPL balls, Narine conceded barely 6.23 rpo (+2.19 Phase ΔEcon) and won his matchup in 77.3% of matches.

• **Jasprit Bumrah (#6 Bowler | PRI: 79.12)**: The greatest death bowler alive. 73.9% of his pressure deliveries were bowled in overs 16–20. Bumrah conceded only 7.32 rpo against a death par of 9.31 (+1.99 runs saved/over) with +248.2 Total BVA.

5. The Other Side of the Coin: Who Chokes Under Pressure?

The data clearly identifies players who retreat into a shell or blow out when the heat rises:

• **Batters Who Stall (Negative ΔSR)**: Curtis Campher (SR drops to 115.61, -5.04 ΔSR, -46.3 RVA), Washington Sundar (SR drops to 136.73, -3.80 ΔSR, -50.8 RVA), Deepak Hooda (SR 130.71, -2.87 ΔSR), Jimmy Neesham (SR 132.46, -2.78 ΔSR), and Wanindu Hasaranga (pressure average collapses to 9.17, -137.9 RVA).

• **Bowlers Who Bleed Runs (Negative Phase ΔEcon)**: Andre Russell (concedes 11.69 rpo under pressure, -2.29 Phase ΔEcon, -128.8 BVA), Nitish Reddy (10.52 rpo, -1.60 Phase ΔEcon), Tushar Deshpande (11.43 rpo, -1.39 Phase ΔEcon, -87.8 BVA), and Mukesh Kumar (11.15 rpo, -1.33 Phase ΔEcon, -135.9 BVA).

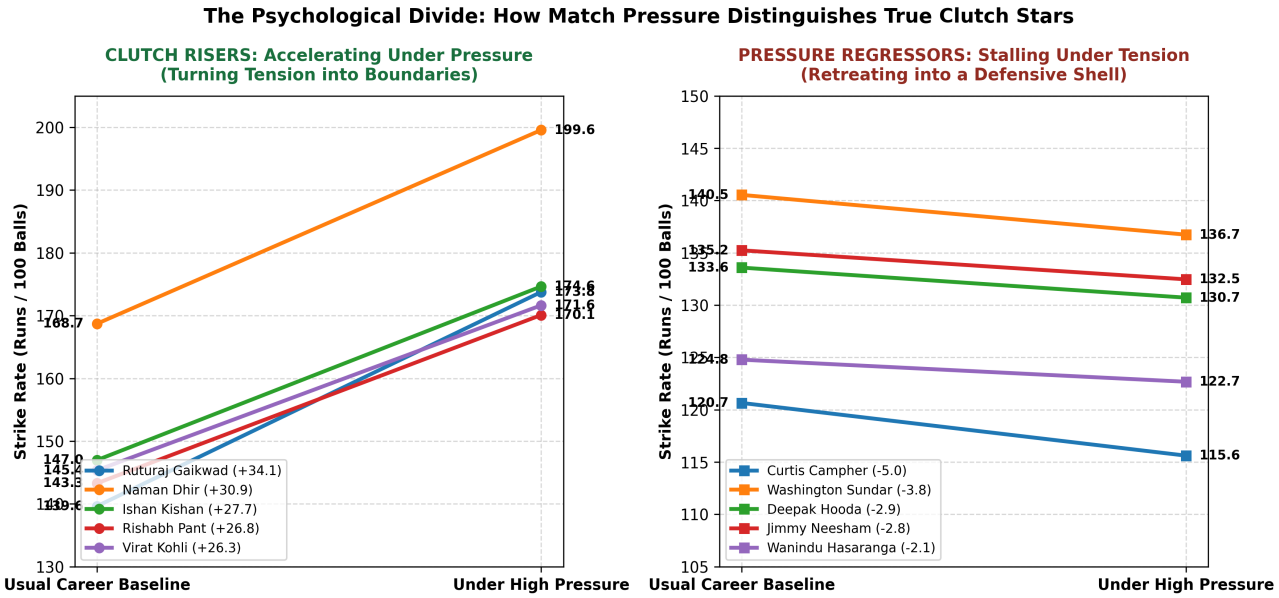


Figure 7: The Psychological Divide: Comparing Clutch Risers (Accelerators) vs Pressure Regressors (Stallers)

6. Strategic Recommendations for Rajasthan Royals

1. **Target Undervalued Clutch Gems**: Naman Dhir and Shashank Singh provide world-class clutch execution at a fraction of superstar auction costs.
2. **Protect High-Variance Bowlers**: Bowlers who blow out at the death (Russell, Deshpande, Mukesh) must be front-loaded in overs 1–14; overs 17–20 must be reserved exclusively for elite run suppressors (Bumrah, Shaheen, Narine).
3. **Pair Composure with Explosion**: In tight chases, pairing an anchor who cuts dot balls (Kohli, Gaikwad) with an explosive boundary striker (David, Shashank) maximizes win expectancy by preventing both collapses and rate asphyxiation.